📘 Day 21 Notes: LLM Model Size Tradeoffs (7B vs 13B vs 70B vs GPT-4)

🚀 Introduction

When we say:

- Mistral 7B
- LLaMA 13B
- LLaMA 70B

👉 The number (7B, 13B, 70B) represents:

🧠 Number of Parameters

🔷 1. What are Parameters in LLMs?

This is the MOST IMPORTANT concept.

✅ Simple Definition

Parameters are:

The internal learned values of the model that store patterns, relationships, grammar, logic, and knowledge learned during training.

They are basically:

- numerical weights
- mathematical values
- learned connections

🧠 Real-Life Analogy (VERY IMPORTANT)

Imagine a human brain.

When a child grows:

- learns language
- understands emotions
- remembers facts
- solves problems

👉 The brain creates stronger neural connections.

In LLMs:

- Parameters = artificial neural connections

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🔍 Another Analogy

Think of parameters like:

📚 "Memory + Experience

More parameters =

- more patterns learned
- better understanding
- deeper reasoning capability

🔷 2. How Parameters Wor

LLMs are neural networks.

Each neuron has:

- weights
- biases

These values are adjusted during t

🔷 3. What Does "7 Billion Parameters" Mean?

When we say:

🧠 Mistral 7B

It means:

- the model contains

👉 7 billion trainable numerical values

These values collectively store:

- grammar
- language patterns
- reasoning patterns
- facts
- coding knowledge
- relationships

🔷 4. Why More Parameters = More Knowledge?

🧠 Simple Analogy

Imagine:

👦 Student A

- studied 10 books

👨‍🎓 Student B

- studied 10,000 books

Who has:

- broader knowledge?
- better understanding?
- better answer quality?

📌 Same in LLMs

Small model:

- limited capacity
- stores fewer patterns

Large model:

- stores far more relationships and patterns

🔍 Example

🟢 Small Model (7B)

Question:

Plain text

```
Why is the sky blue?
```

Answer:

Plain text

```
Because sunlight scatters.
```

Simple but shallow.

🔴 Large Model (70B)

Answer:

Plain text

```
The sky appears blue becaus
```

More detailed.
More scientifically accurate.

🔷 5. What is "Reasoning" in LLMs?

This is VERY misunderstood.

❌ Wrong Understanding

LLMs do NOT truly "think" like humans.

✅ Correct Understanding

Reasoning means:

> Ability to combine learned patterns to solve multi-step problems

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🧠 Example

Question:

Plain text

```
A train leaves Delhi at 5 PM at 60 km/h...
```

The model must:

1. understand problem
2. extract numbers
3. apply logic
4. generate final answer

That multi-step capability =

👉 reasoning

🔷 6. Why Bigger Models Have Better Reasoning

Because larger models:

- see more examples during training
- learn more logical patterns
- capture deeper relationships

🧠 Real-Life Analogy

👦 Child

Can answer:

```
Plain text

2 + 2 = 4
```

👨‍🔬 Scientist

Can solve:

- calculus
- physics
- advanced logic

Why?
👉 More learning + more experience.

📌 Same in LLMs

70B models:

- saw more complex patterns
- learned deeper logical relationships

🔷 7. Important Truth: Bigger ≠ Always Better

THIS IS INTERVIEW GOLD.

❌ Common Mistake

People think:

```
Plain text

bigger model = always best
```

Wrong.

✅ Real Industry Thinking

The best model is:

👉 the most optimal model

not the biggest.

🔍 Example

Suppose your app:

- only summarizes emails

Do you need GPT-4-level reasoning?

❌ No.

A small 7B model is enough.

💰 Why?

Because large models:

- cost more
- require GPUs
- consume memory
- increase latency

🔷 8. Understanding Latency

Latency =

⏳ response time

🟢 Small Models

- respond faster

🔴 Large Models

- slower due to huge computations

🔍 Example

7B:

```
response in 1 second
```

70B:

```
response in 8 seconds
```

🔷 9. GPU Memory Requirement

Large models require:

- huge VRAM
- expensive GPUs

🔍 Example

Model	Approx VRAM
7B	8–16 GB
13B	24 GB
70B	80+ GB

🧠 Why?

More parameters =

more mathematical computations.

🔷 10. Real Production Example

🏢 Example 1: Customer Support Chatbot

Needs:

- fast replies
- low cost

Use:

✅ 7B model

🏢 Example 2: Legal AI Assistant

Needs:

- complex reasoning
- accuracy

Use:

✅ GPT-4 / 70B

🔹 11. Fine-Tuning vs Large Models

Interesting industry trend:

Instead of:

❌ using huge model

companies often:

✅ fine-tune smaller models

Why?

Because:

- cheaper
- faster
- specialized

🔍 Example

7B medical model fine-tuned on hosp
can outperform generic 70B model ir

🔹 13. Hallucination vs Model Size

Hallucination =
model generates false information.

🟢 Small Models

More hallucinations sometimes.

🔴 Larger Models

Usually better factual consistency.

But hallucinations still exist.

🔹 14. Context Window vs Parameters

IMPORTANT:
People confuse these.

❌ Parameters

= model intelligence

✅ Context Window

= how much text model remembers at once

🔍 Example

GPT-4:

- huge reasoning
- huge context

🔷 15. Mixture of Experts (MoE)

Modern trick used in:

- Mixtral
- GPT-4 architecture rumors

🧠 Idea

Instead of activating whole model,
only activate relevant “experts”.

🚀 Benefit

- cheaper
- faster
- scalable

🔷 16. Rule of Thumb

🟢 Use Small Models When:

- speed matters
- budget matters
- local deployment needed
- simple tasks

🔴 Use Large Models When:

- reasoning matters
- coding tasks
- legal/finance
- deep analysis